Quantum Mechanics I — Exercise Sheet 2

A driven qubit

Variant: Questions only

Conventions, as in Lecture 2: the circular drive $H(t) = \frac{\hbar\nu_0}{2}\sigma_z + \frac{\hbar\Omega}{2}\big(\cos(\nu t)\,\sigma_x + \sin(\nu t)\,\sigma_y\big)$ with $\nu_0, \Omega, \nu > 0$; $R(t) = e^{-i\nu t\sigma_z/2}$, $\psi_t = R(t)\tilde{\psi}_t$; $\Delta = \nu_0 - \nu$; in the frame the Hamiltonian is $\tilde{H} = \frac{\hbar}{2}(\Delta\sigma_z + \Omega\sigma_x)$, and the frame Bloch vector $\tilde{\mathbf{r}}$ turns about $(\Omega, 0, \Delta)/\tilde{\Omega}$ at the rate $\tilde{\Omega} = \sqrt{\Omega^2 + \Delta^2}$.

Exercise 1 (another starting point). On resonance, a spin starts in the state $|{+y}\rangle$, with Bloch vector $\mathbf{e}_y$. Find $\tilde{\mathbf{r}}(t)$ and the probability of reading $|{\downarrow}\rangle$ at the time t. When is it first 1? What happens if the spin starts in $|{+x}\rangle$ instead?

Exercise 2 (a π-pulse inverts every state). Show that the π-pulse $-i\sigma_x$ sends every frame Bloch vector $(\tilde{r}_x, \tilde{r}_y, \tilde{r}_z)$ to $(\tilde{r}_x, -\tilde{r}_y, -\tilde{r}_z)$. What does it do to the probabilities of $|{\uparrow}\rangle$ and $|{\downarrow}\rangle$?

Exercise 3 (why the rotating frame is needed). For a constant H the solution is $e^{-itH/\hbar}\psi_0$. A natural guess for the drive is $\psi_T = \exp\big(-\frac{i}{\hbar}\int_0^T H(t)\,dt\big)\psi_0$. (i) Compute $\int_0^T H(t)\,dt$. (ii) On resonance, $\nu = \nu_0$, what does the guess predict for the probability of $|{\downarrow}\rangle$ at $T = 2\pi/\nu$, from $|{\uparrow}\rangle$? What is the true value? Evaluate both for $\Omega/\nu = 0.1$.

Exercise 4 (Ramsey's two pulses). A spin prepared up receives a $\pi/2$-pulse, then evolves for a time T with the drive switched off, then receives a second $\pi/2$-pulse whose drive keeps the phase of the first, $\cos(\nu t)$ and $\sin(\nu t)$ continuing in t. Assume that the pulses are so short that the detuning Δ can be neglected during them, so that each turns $\tilde{\mathbf{r}}$ about $\mathbf{e}_x$ through $\pi/2$. (i) Show that with the drive off the frame Hamiltonian is $\frac{\hbar\Delta}{2}\sigma_z$. (ii) Follow $\tilde{\mathbf{r}}$ through the three steps and find the probability of $|{\downarrow}\rangle$ at the end. (iii) For $T = 1$ ms, how far apart in $\nu/2\pi$ are the fringes, and what is the full width at half maximum of the central one?

Exercise 5 (the shift of a linear drive, an estimate ⋆). In Remark 2.4, a linear drive adds to the circular one the counter-rotating term $\frac{\hbar\Omega}{2}\big(\cos(\nu t)\,\sigma_x - \sin(\nu t)\,\sigma_y\big)$. (i) Show that in the frame $\psi = e^{+i\nu t\sigma_z/2}\chi$, which turns with the counter-rotating field, that term becomes $\frac{\hbar\Omega}{2}\sigma_x$ and the static term becomes $\frac{\hbar}{2}(\nu_0 + \nu)\sigma_z$. (ii) Ignore the co-rotating term in this frame. Then the spin precesses about $(\Omega, 0, D)/\sqrt{D^2 + \Omega^2}$, $D = \nu_0 + \nu$, at the rate $\sqrt{D^2 + \Omega^2}$. Show that for $\Omega \ll D$ this rate exceeds D by about $\Omega^2/(2D)$. (iii) Near resonance $D \approx 2\nu_0$. Conclude that the counter-rotating term raises the spin's own frequency, and so the resonance, by about $\Omega^2/(4\nu_0)$, and compare with a numerical solution for $\Omega/\nu_0 = 0.08$. Over one period $T = 2\pi/\nu$ of the drive, the evolution operator is computed numerically. The angle between its two eigenvalues, divided by T, is the rate at which the spin effectively turns, and the resonance is taken where this rate is smallest. This puts the resonance at $\nu/\nu_0 = 1.0016$, to four decimal places.